

Existence and Localization of a Limit Cycle in a Class of Benchmark Biomolecular Oscillators

Sidhanta Mohanty, Shaunak Sen

Abstract

Oscillatory behaviour is important in multiple biological contexts. However, the inherent nonlinearity and high dimensionality of mathematical models in biology makes proving the existence and the localization of limit cycle oscillations challenging. Here, we provided an elementary proof for the existence and a method for rigorously localizing the oscillatory solutions in a class of benchmark biomolecular oscillators. To construct the proof, we used a geometric approach based on Brouwer's Fixed Point theorem. We constructed a toroidal-like manifold within a positively invariant set by removing the hypervolume containing the fixed point and the trajectories converging to it along its stable manifold. We showed that the vector field describing the system dynamics maps a cross section of the toroidal-like manifold onto itself. The existence of a limit cycle solution in this manifold was guaranteed by Brouwer's Fixed Point theorem. For different sets of initial conditions in these cross-sections, we used an interval-based Reachability Analysis to localize the oscillatory behaviour. These results add a simple and elegant approach to demonstrating the existence of limit cycles in biomolecular systems as well as a method for rigorous localization of the region of existence.

Index Terms

Limit Cycle, Cyclic Gene Regulatory Network, Brouwer's Fixed Point Theorem, Reachability Analysis.

I. INTRODUCTION

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II. INTRODUCTION

The Poincaré-Bendixson theorem provides a way to demonstrate the existence of limit cycle oscillations in two-dimensional systems [1]. The basic idea is to construct a positively invariant set without steady states so that the trajectories converge to the limit cycle. An explicit solution of the limit cycle is

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not needed. This theorem is not applicable in dimensions greater than two because of the possibility of more complex attractors and chaotic behaviour. Limit cycles in systems of dimension greater than two, however, are typically found in biological systems, where they underlie crucial control processes such as circadian rhythms and the cell cycle [2]. For example, the famous Goodwin oscillator has a negative feedback loop wrapping a chain of intermediate biomolecules so that the overall length of the chain is greater than or equal to three [3]–[5]. Another canonical example is a ring oscillator with an odd number (≥ 3) of genes that sequentially repress each other [6]. The experimental demonstration of the repressilator, a ring oscillator with three genes, heralded a new era for synthetic biology [7]. Therefore, methods to establish the existence and rigorous enclosure of limit cycles in higher-dimensional systems are needed.

While various approaches have been used to prove the existence of limit cycles in biological systems of dimensions greater than or equal to three, the region of existence is relatively large and conservative. Like the Poincare-Bendixson theorem, these approaches seek to establish the existence of a limit cycle without an explicit analytical solution. One approach is to analyze biological oscillators in a Boolean framework using kinetic logic models [6], [8]. A geometric approach based on Brouwer’s fixed point theorem was used to prove the existence of a limit cycle in the three-dimensional Goodwin oscillator [9], which we recently adapted for the repressilator [10]. This approach was subsequently generalized to an n -dimensional Goodwin oscillator [11]. This approach provides concrete insight into the limit cycle behaviour, but has been referred to as ‘elaborate’ [12] and ‘laborious’ [13]. An n -dimensional version of the Poincare-Bendixson Theorem was developed by for monotone cyclic feedback systems [14]. The above ideas were used to address the existence and stability in ring oscillators and their variants [12], [15]. An explicit characterization of biological parameter sets for the existence of oscillations for the ring oscillators was also reported [16]–[18]. Recent work has expanded on the notion of 2-cooperativity to prove the existence of limit cycles with wide applicability in biology [19]–[21]. Typical estimates of the region of existence of the limit cycle obtained using the above approaches are conservative in nature, and rigorous tighter enclosures are desirable.

In this work, we localized the region of existence of the limit cycle in a benchmark biomolecular oscillators using the Brouwer’s Fixed Point theorem and interval-based Reachability Analysis. We provided an alternative, elementary proof for the existence of a limit cycle in cyclic gene regulatory network with odd number of genes. For the existence proof, we constructed a positively invariant set for oscillator dynamics by analyzing the direction of the vector field at the outer facets of a hypercube. We constructed a torus-like structure by removing a hypervolume associated with the unstable steady state and the trajectories approaching it along its stable manifold. We showed that a cross-section of

this torus-like region maps onto itself, thus proving the existence of a limit cycle. For different sets of initial conditions on these cross-sections, we rigorously evolved the network dynamics using Interval Analysis to localize the limit cycle.

III. BROUWER'S FIXED POINT THEOREM

Brouwer's Fixed Point theorem can be used to ascertain the presence of a fixed point. The Brouwer's Fixed Point theorem states:

Theorem III.1 ([22]). *Let $S \subset \mathbb{R}^n$ be convex and compact, and let $T : S \rightarrow S$ be continuous. Then T has a fixed point in S .*

IV. EXISTENCE OF LIMIT CYCLE

In this section, an elementary proof for the existence of a limit cycle in m -gene cyclic regulatory network based on Brouwer's Fixed Point theorem is presented.

Let $x_1, x_2, \dots, x_m$ represent the proteins of the m -gene cyclic regulatory network. The differential equations describing the dynamics of the network are given in (1). A symmetric model with only the protein dynamics, ignoring mRNA dynamics, is chosen for simplicity.

$$\begin{aligned} \frac{dx_1}{dt} &= F(x_m) - \gamma x_1, \\ \frac{dx_2}{dt} &= F(x_1) - \gamma x_2, \\ &\vdots \\ \frac{dx_m}{dt} &= F(x_{m-1}) - \gamma x_m, \end{aligned} \tag{1}$$

where $x_1, x_2, \dots, x_m$ are protein concentrations. α, γ, k , and n are positive constants, $F(p) = \frac{\alpha}{1 + (\frac{p}{k})^n}$ is the Hill function, n is the Hill coefficient, and m is an odd number.

The steady state of the cyclic regulatory network in (1) is unique. The uniqueness can be demonstrated graphically [2]. Denote the steady state as $(x_{10}, x_{20}, \dots, x_{m0})$. So,

$$x_{10} = \underbrace{\frac{1}{\gamma} F \left(\frac{1}{\gamma} F \dots \left(\frac{1}{\gamma} F(x_{10}) \right) \dots \right)}_{m\text{-times}}.$$

The stability of the steady state can be assessed through the eigenvalues of the Jacobian of the linearization of the system in (1) at the steady state $(x_0, x_0, \dots, x_0)$. The eigenvalues of the Jacobian are:

$$\lambda_{1,2,\dots,m} = -\gamma + \rho \left[\cos \left(\frac{2p+1}{m} \pi \right) + j \sin \left(\frac{2p+1}{m} \pi \right) \right]. \tag{2}$$

where $p = 0, 1, \dots, m-1$ and $\rho = -F'(x_0)$. An eigenvalue is real and negative, $\lambda_1 = -(\gamma + \rho)$. The remaining eigenvalues are complex in nature. For certain values of p , the cosine component in the real part of the eigenvalues become negative. This implies that the corresponding eigenvalues will certainly have negative real parts. We used the instability of the steady state to prove the following sufficient condition for the existence of a limit cycle.

Theorem IV.1. *There exists a limit cycle solution to the cyclic gene regulatory network in (1) if $\gamma < \max_{p \in \mathcal{P}_+} \left\{ \rho \cos \left(\frac{2p+1}{m} \pi \right) \right\}$, where $\mathcal{P}_+ = \left\{ p \in [0, m-1] : \cos \left(\frac{2p+1}{m} \pi \right) > 0 \right\}$.*

Proof. For certain values of p in (2), the cosine component in the real part of the eigenvalues becomes negative. This implies that the corresponding eigenvalues will certainly have a negative real part. This contributes to the possible stability of the steady state. However, the instability of the steady state implies that nearby trajectories are repelled from the steady state and may converge to a closed invariant set, such as a limit cycle. To ensure that the steady state is unstable, at least one of the complex conjugate pair of eigenvalues should have a positive real part. For the complex eigenvalues to have a positive real part

$$-\gamma + \rho \cos \left(\frac{2p+1}{m} \pi \right) > 0 \quad \forall p \in \mathcal{P}_+,$$

where, $\mathcal{P}_+ = \left\{ p \in [0, m-1] : \cos \left(\frac{2p+1}{m} \pi \right) > 0 \right\}$. So, to ensure atleast one of the complex conjugate pair of eigenvalues have positive real part

$$\gamma < \max_{p \in \mathcal{P}_+} \left\{ \rho \cos \left(\frac{2p+1}{m} \pi \right) \right\}. \quad (3)$$

Upon establishing the instability of the steady state, a geometric approach is used to prove the existence of a limit cycle within the hypercube $0 \leq x_1, x_2, \dots, x_m \leq \frac{\alpha}{\gamma}$.

Construction of an Invariant Box: To demonstrate the existence of limit cycle in the m -gene cyclic regulatory network described by (1), we showed that the hypercube is a positively invariant set. This is done by computing the inner product between the vector field characterized by the system dynamics within the hypercube and the normal to the outer facets of the hypercube,

$$\Xi = \langle \vec{G}, \hat{n} \rangle,$$

where G is the vector field that describes the dynamics of the m -gene cyclic regulatory network and $\hat{n}$ is the unit vector normal to the facet. We considered the unit normal vectors for the different facets

of the hypercube directed inward. The value of Ξ determines the direction of the flow of the vector field through the facets. So,

$$\Xi \begin{cases} > 0, \text{ the vector field flows through the facet in the direction of } \hat{n}, \\ = 0, \text{ the vector field does not flow through the facet,} \\ < 0, \text{ the vector field flows through the facet in the direction opposite of } \hat{n}. \end{cases}$$

The vector field described by the system dynamics in (1) is

$$\vec{G}(x_1, x_2, x_3, \dots, x_m) = [F(x_m) - \gamma x_1] \hat{x}_1 + [F(x_1) - \gamma x_2] \hat{x}_2 + [F(x_2) - \gamma x_3] \hat{x}_3 \\ + \dots + [F(x_{m-2}) - \gamma x_{m-1}] \hat{x}_{m-1} + [F(x_{m-1}) - \gamma x_m] \hat{x}_m,$$

where $\hat{x}_i$ (for $i = 1, 2, \dots, m$) are the unit vectors in the direction of x_i axis. We calculated the inner product (Ξ) of the vector field with unit normal vector to the different facets. The normal unit vector is always considered to be directed into the hypercube. The outer facets of the hypercube are $x_1 = 0$, $x_1 = \frac{\alpha}{\gamma}$, $x_2 = 0$, $x_2 = \frac{\alpha}{\gamma}$, $\dots$, $x_m = 0$, and $x_m = \frac{\alpha}{\gamma}$. We computed the inner product between the vector field characterized by the system dynamics within the hypercube and the normal to the outer facets of the hypercube.

Table I: Value of the inner product between the vector field and the normal to the outer facets of the hypercube.

Facet	Unit normal Vector, $\hat{n}$	Inner Product, $\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_i = 0$	$\hat{x}_i$	$\frac{\alpha}{1 + \left(\frac{x_{i-1}}{k}\right)^n}$	+ve
$x_i = \frac{\alpha}{\gamma}$	$-\hat{x}_i$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_{i-1}}{k}\right)^n}$	+ve
for $i = 1, 2, 3, \dots, m$. As it is a cyclic network, $x_0 = x_m$.			

The inner product between the vector field characterized by the system dynamics within the hypercube and the normal to the outer faces of the hypercube is given in Table I. For protein concentrations between 0 and $\frac{\alpha}{\gamma}$, the value of Ξ is always positive. This implies that the vector field flows through the hyperplane into the hypercube. Thus, the vector field flowing through all the outer surfaces is always directed into the hypercube, making the hypercube a positively invariant box.

Construction of an m - dimensional Toroidal-like Manifold: Trajectories approach the unique steady

state along the path defined by the eigenvector of the negative eigenvalue. We calculated the eigenvector for the real negative eigenvalue. Let the eigenvector be $\begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 & \cdots & \hat{x}_m \end{bmatrix}^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & \cdots & -\rho \\ -\rho & -\gamma & 0 & \cdots & 0 \\ 0 & -\rho & -\gamma & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\gamma \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \vdots \\ \hat{x}_m \end{bmatrix} = -(\gamma + \rho) \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \vdots \\ \hat{x}_m \end{bmatrix},$$

$$\implies \hat{x}_m = \hat{x}_1, \hat{x}_1 = \hat{x}_2, \dots, \hat{x}_{m-1} = \hat{x}_m,$$

$$\implies \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \vdots \\ \hat{x}_m \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}.$$

In the direction of the eigenvector $\begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 & \cdots & \hat{x}_m \end{bmatrix}^T$, we cut out a hypervolume from the given hypercube such that the steady state and the trajectories approaching it along this singular path are eliminated. This hypervolume may be visualized as a cyclinder with a small radius $\varepsilon > 0$ along the eigenvector.

For some values of p , there is a possibility that some of these complex conjugate eigenvalues may have negative real part. So, we calculated the eigenvectors associated with such complex eigenvalues and removed the corresponding hypervolume along it. Let the eigenvector associated with such complex eigenvalues be $\begin{bmatrix} \tilde{x}_1 & \tilde{x}_2 & \tilde{x}_3 & \cdots & \tilde{x}_m \end{bmatrix}^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & \cdots & -\rho \\ -\rho & -\gamma & 0 & \cdots & 0 \\ 0 & -\rho & -\gamma & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & -\gamma \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_m \end{bmatrix} = (-\gamma + \rho f(p, m)) \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_m \end{bmatrix},$$

where $f(p, m) = \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right]$ and we have

$$\begin{aligned}\tilde{x}_m &= \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right] \tilde{x}_1, \\ \tilde{x}_1 &= \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right] \tilde{x}_2, \\ &\vdots \\ \tilde{x}_{m-1} &= \left[\cos\left(\frac{2p+1}{m}\pi\right) + j \sin\left(\frac{2p+1}{m}\pi\right) \right] \tilde{x}_m.\end{aligned}\tag{4}$$

Using De Moivre's theorem on (4), we get the eigenvector as:

$$\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \vdots \\ \tilde{x}_{m-1} \\ \tilde{x}_m \end{bmatrix} = \begin{bmatrix} 1 \\ \cos\left(\frac{(m-1)(2p+1)}{m}\pi\right) + j \sin\left(\frac{(m-1)(2p+1)}{m}\pi\right) \\ -\cos\left(\frac{(m-2)(2p+1)}{m}\pi\right) - j \sin\left(\frac{(m-2)(2p+1)}{m}\pi\right) \\ \vdots \\ \cos\left(\frac{2(2p+1)}{m}\pi\right) + j \sin\left(\frac{2(2p+1)}{m}\pi\right) \\ -\cos\left(\frac{2p+1}{m}\pi\right) - j \sin\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}.$$

Although complex eigenvectors are not elements of the real space, their real and imaginary components reside within it. We divided the above eigenvector into its real and imaginary parts. These are the eigenvectors associated with a pair of complex conjugate eigenvalues,

$$v_{real} = \begin{bmatrix} 1 \\ \cos\left(\frac{(m-1)(2p+1)}{m}\pi\right) \\ -\cos\left(\frac{(m-2)(2p+1)}{m}\pi\right) \\ \vdots \\ \cos\left(\frac{2(2p+1)}{m}\pi\right) \\ -\cos\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}, v_{im} = \begin{bmatrix} 0 \\ \sin\left(\frac{(m-1)(2p+1)}{m}\pi\right) \\ -\sin\left(\frac{(m-2)(2p+1)}{m}\pi\right) \\ \vdots \\ \sin\left(\frac{2(2p+1)}{m}\pi\right) \\ -\sin\left(\frac{2p+1}{m}\pi\right) \end{bmatrix}.$$

Because some of these complex conjugate eigenvalues could have a negative real part, we removed the hypervolume associated with the linear combination of v_{real} and v_{im} .

So, we obtained an m -dimensional toroidal-like manifold without the steady state. All the trajectories that enter this manifold stay inside because the hypercube is a positively invariant set and there are not trajectories that approach the steady state.

Continuous Map of a Cross-Section Onto Itself: We divided the hypercube into 2^m smaller hyperrectangles using the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$). Further, we computed the flow that traverses every facet within the smaller hyperrectangles.

Table II: Value of the inner product between the vector field and the normal to the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$).

Hyperplane	$\hat{n}$	$\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_i = x_0$	$\hat{x}_i$	$\frac{\alpha}{1 + \left(\frac{x_i - 1}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_{i-1} < x_0, \\ -\text{ve, for } x_{i-1} > x_0. \end{cases}$
for $i = 1, 2, 3, \dots, m$. As it is a cyclic network, $x_0 = x_m$.			

The flow of the vector field through the outer facets of the hypercube obtained in Table I, combined with the flow through the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$) obtained in Table II, is used to determine the effective flow from one smaller hyperrectangle to its adjacent ones. We considered one of the smaller hyperrectangles, $0 \leq x_1 \leq x_0, 0 \leq x_2 \leq x_0, \dots, 0 \leq x_m \leq x_0$, as Box 1. From Table I, we have the flow of the vector field from the facets $x_i = 0$ (for $i = 1, 2, 3, \dots, m$) into Box 1. From Table II, we have the flow of the vector field from the hyperplanes $x_i = x_0$ (for $i = 1, 2, 3, \dots, m$) out of Box 1 and into the adjacent boxes. Similarly, we calculated the effective flow of the vector field for the rest of the boxes.

Upon inspection of the effective flow from one hyperrectangle to another, we observed that the vector field forms a closed path within certain hyperrectangles. This showed that a cross section of these smaller hyperrectangles mapped onto itself. By Brouwer's fixed point theorem, there exists a fixed point, which is a limit cycle for the system under consideration. The above construction also provides a rigorous bound on the region within the toroidal-like manifold where the limit cycle exists. $\square$

To summarize, we obtained the region in the hypercube where the limit cycle exists. This is done by constructing a toroidal-like manifold within a positively invariant set obtained by eliminating a hypervolume containing the steady state and the trajectories approaching it. Finally, we used Brouwer's Fixed Point theorem on a cross section of this manifold, which map onto itself, to show the existence of limit cycle.

V. CASE STUDY

A. 5-gene cyclic regulatory network

The differential equations describing the dynamics of the 5-gene cyclic regulatory network are given in (5).

$$\begin{aligned}\frac{dx_1}{dt} &= F(x_5) - \gamma x_1, \\ \frac{dx_2}{dt} &= F(x_1) - \gamma x_2, \\ \frac{dx_3}{dt} &= F(x_2) - \gamma x_3, \\ \frac{dx_4}{dt} &= F(x_3) - \gamma x_4, \\ \frac{dx_5}{dt} &= F(x_4) - \gamma x_5,\end{aligned}\tag{5}$$

where x_1, x_2, x_3, x_4 , and x_5 are the protein concentrations with $0 \leq x_1 \leq \frac{\alpha}{\gamma}$, $0 \leq x_2 \leq \frac{\alpha}{\gamma}$, $0 \leq x_3 \leq \frac{\alpha}{\gamma}$, $0 \leq x_4 \leq \frac{\alpha}{\gamma}$, and $0 \leq x_5 \leq \frac{\alpha}{\gamma}$. α, γ, k , and n are positive constants. $F(p) = \frac{\alpha}{1+(\frac{p}{k})^n}$ is the Hill function and n is the Hill coefficient. The eigenvalues of the Jacobian after the linearization of the system in (5) at the steady state $(x_0, x_0, x_0, x_0, x_0)$ is

$$\lambda_{1,2,\dots,5} = -(\gamma + \rho), -\gamma + \rho(-0.31 \pm j0.95), -\gamma + \rho(0.81 \pm j0.59).\tag{6}$$

The 5-gene regulatory network has a negative real eigenvalue. Along with it, the complex conjugate eigenvalues could also have a negative real part. Guided by Theorem IV.1, we illustrated the existence of the limit cycle by considering the case where one pair of the complex conjugate eigenvalues has a positive real part.

1) *Construction of an invariant box:* We calculated the flow of the vector field through the different facets of the hypercube $0 \leq x_1, x_2, x_3, x_4, x_5 \leq \frac{\alpha}{\gamma}$. This is calculated along the unit normal vector of the various facets directed into the hypercube. Using the value of Ξ , we determined the flow of the vector field through the various facets and checked if the hypercube constituted a positively invariant set.

The vector field described by the system dynamics in (5) is

$$\begin{aligned}\vec{G}(x_1, x_2, x_3, x_4, x_5) &= [F(x_5) - \gamma x_1] \hat{x}_1 + [F(x_1) - \gamma x_2] \hat{x}_2 + [F(x_2) - \gamma x_3] \hat{x}_3 \\ &\quad + [F(x_3) - \gamma x_4] \hat{x}_4 + [F(x_4) - \gamma x_5] \hat{x}_5,\end{aligned}$$

where $\hat{x}_i$ (for $i = 1, 2, \dots, 5$) are the unit vectors in the direction of x_i axis. We calculated the value of Ξ between the vector field and the unit normal vector to the different facets. The unit normal vector is always considered to be directed into the hypercube.

Table III: Value of the inner product between the vector field and the normal to the outer facets of the hypercube.

Facet	Unit normal Vector, $\hat{n}$	Inner Product, $\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_1 = 0$	$\hat{x}_1$	$\frac{\alpha}{1 + \left(\frac{x_5}{k}\right)^n}$	+ve
$x_1 = \frac{\alpha}{\gamma}$	$-\hat{x}_1$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_5}{k}\right)^n}$	+ve
$x_2 = 0$	$\hat{x}_2$	$\frac{\alpha}{1 + \left(\frac{x_1}{k}\right)^n}$	+ve
$x_2 = \frac{\alpha}{\gamma}$	$-\hat{x}_2$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_1}{k}\right)^n}$	+ve
$x_3 = 0$	$\hat{x}_3$	$\frac{\alpha}{1 + \left(\frac{x_2}{k}\right)^n}$	+ve
$x_3 = \frac{\alpha}{\gamma}$	$-\hat{x}_3$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_2}{k}\right)^n}$	+ve
$x_4 = 0$	$\hat{x}_4$	$\frac{\alpha}{1 + \left(\frac{x_3}{k}\right)^n}$	+ve
$x_4 = \frac{\alpha}{\gamma}$	$-\hat{x}_4$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_3}{k}\right)^n}$	+ve
$x_5 = 0$	$\hat{x}_5$	$\frac{\alpha}{1 + \left(\frac{x_4}{k}\right)^n}$	+ve
$x_5 = \frac{\alpha}{\gamma}$	$-\hat{x}_5$	$\alpha - \frac{\alpha}{1 + \left(\frac{x_4}{k}\right)^n}$	+ve

The value of the inner product between the vector field and the normal to the facets directed into the hypercube is given in Table III. The values of Ξ are positive for every outer facet of the hypercube. This indicated that the flow of the vector field described by the system dynamics is always directed inward and stays inside the hypercube, making it a positively invariant set.

2) *Construction of a 5– dimensional Toroidal-like Manifold:* The existence of a unique steady state in the given hypercube indicated that certain trajectories approach the steady state along the singular path defined by the eigenvector of the negative eigenvalue. So, we calculated the eigenvector for the real negative eigenvalue. Let the eigenvector be $\begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 & \hat{x}_4 & \hat{x}_5 \end{bmatrix}^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & 0 & -\rho \\ -\rho & -\gamma & 0 & 0 & 0 \\ 0 & -\rho & -\gamma & 0 & 0 \\ 0 & 0 & -\rho & -\gamma & 0 \\ 0 & 0 & 0 & -\rho & -\gamma \end{bmatrix} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \\ \hat{x}_5 \end{bmatrix} = -(\gamma + \rho) \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \\ \hat{x}_5 \end{bmatrix}. \quad (7)$$

From (7), the eigenvector is found to be

$$\begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \\ \hat{x}_3 \\ \hat{x}_4 \\ \hat{x}_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}.$$

In the direction of the eigenvector $\begin{bmatrix} \hat{x}_1 & \hat{x}_2 & \hat{x}_3 & \hat{x}_4 & \hat{x}_5 \end{bmatrix}^T$, we cut out a hypervolume from the given hypercube, such that the steady state and the trajectories approaching it along the singular path are eliminated.

The regulatory network also has a pair of complex conjugate eigenvalues with negative real part. So, we calculated the eigenvectors associated with these complex eigenvalues and remove a hypervolume along it. Let the eigenvector associated with one of these complex eigenvalues be $\begin{bmatrix} \tilde{x}_1 & \tilde{x}_2 & \tilde{x}_3 & \tilde{x}_4 & \tilde{x}_5 \end{bmatrix}^T$. So,

$$\begin{bmatrix} -\gamma & 0 & 0 & 0 & -\rho \\ -\rho & -\gamma & 0 & 0 & 0 \\ 0 & -\rho & -\gamma & 0 & 0 \\ 0 & 0 & -\rho & -\gamma & 0 \\ 0 & 0 & 0 & -\rho & -\gamma \end{bmatrix} \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \end{bmatrix} = -\gamma + \rho(-0.31 + j0.95) \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \end{bmatrix}.$$

Using the relation in (4) and subsequently using the De Moivre's theorem, we get the eigenvector as:

$$\begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \\ \tilde{x}_4 \\ \tilde{x}_5 \end{bmatrix} = \begin{bmatrix} 1 \\ 0.31 + j0.95 \\ -0.81 + j0.59 \\ -0.81 - j0.59 \\ 0.31 - j0.95 \end{bmatrix}.$$

We divide the above eigenvector into its real and imaginary parts. These are the eigenvectors associated complex eigenvalue with negative real part.

$$v_{real} = \begin{bmatrix} 1 \\ 0.31 \\ -0.81 \\ -0.81 \\ 0.31 \end{bmatrix}, v_{im} = \begin{bmatrix} 0 \\ 0.95 \\ 0.59 \\ -0.59 \\ -0.95 \end{bmatrix}.$$

The complex conjugate eigenvalues produce conjugate eigenvectors. So, the v_{real} remains same for the complex conjugate pair. v_{im} for the pair of eigenvalues is a conjugate pair. However, the conjugate gives the same eigenvector but in the opposite direction. In the sense of removing a hypervolume, the v_{im} can be considered to be same. So, we also removed the hypervolume along the eigenvectors v_{real} and v_{im} .

So, we obtained a toroidal-like manifold with no steady state. All the trajectories are inside the torus-like structure (because the hypercube is a positively invariant set).

3) *Continuous Map of a Cross-Section Onto Itself*: The hypercube is divided into 2^5 smaller hyperrectangles by using the hyperplanes $x_1 = x_0$, $x_2 = x_0$, $x_3 = x_0$, $x_4 = x_0$, and $x_5 = x_0$. We calculated the value of Ξ between the vector field describing the system dynamics and the unit normal vector through these hyperplanes.

Table IV: Value of the inner product between the vector field and the normal to the hyperplanes $x_i = x_0$ (for $i = 1, 2, \dots, 5$).

Hyperplane	Unit normal Vector, $\hat{n}$	Inner Product, $\Xi = \langle \vec{G}, \hat{n} \rangle$	Value of Ξ
$x_1 = x_0$	$\hat{x}_1$	$\frac{\alpha}{1 + \left(\frac{x_5}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_5 < x_0, \\ -\text{ve, for } x_5 > x_0. \end{cases}$
$x_2 = x_0$	$\hat{x}_2$	$\frac{\alpha}{1 + \left(\frac{x_1}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_1 < x_0, \\ -\text{ve, for } x_1 > x_0. \end{cases}$
$x_3 = x_0$	$\hat{x}_3$	$\frac{\alpha}{1 + \left(\frac{x_2}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_2 < x_0, \\ -\text{ve, for } x_2 > x_0. \end{cases}$
$x_4 = x_0$	$\hat{x}_4$	$\frac{\alpha}{1 + \left(\frac{x_3}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_3 < x_0, \\ -\text{ve, for } x_3 > x_0. \end{cases}$
$x_5 = x_0$	$\hat{x}_5$	$\frac{\alpha}{1 + \left(\frac{x_4}{k}\right)^n} - \gamma x_0$	$\begin{cases} +\text{ve, for } x_4 < x_0, \\ -\text{ve, for } x_4 > x_0. \end{cases}$

The value of the inner product between the vector field and the normal to the hyperplanes $x_1 = x_0$, $x_2 = x_0$, $x_3 = x_0$, $x_4 = x_0$, and $x_5 = x_0$ is given in Table IV. Using the values of Ξ in Table III and Table IV, we determined the direction of flow of the vector field within the smaller boxes. This is summarized in Table V.

Table V: Direction of flow of vector field within the smaller boxes.

Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
1	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 2 $\uparrow$ Box 32 $\leftarrow$ Box 1 $\rightarrow$ Box 29 $\swarrow \quad \searrow$ Box 31 Box 32
2	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 25 $\nearrow$ Box 2 $\rightarrow$ Box 26 $\searrow$ Box 28
3	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 2 $\nearrow$ Box 3 $\rightarrow$ Box 18 $\searrow$ Box 27
4	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 10 $\nearrow$ Box 4 $\rightarrow$ Box 3 $\searrow$ Box 28
5	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 4 $\nearrow$ Box 5 $\rightarrow$ Box 17 $\searrow$ Box 19
6	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 5 $\uparrow$ Box 10 $\leftarrow$ Box 6 $\rightarrow$ Box 7 $\swarrow \quad \searrow$ Box 9 Box 8
7	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 $\nearrow$ Box 7 $\rightarrow$ Box 13 $\searrow$ Box 14
Continued on the next page.		

Continuation of Table V ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
8	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 11 $\nearrow$ Box 8 $\rightarrow$ Box 16 $\searrow$ Box 17
9	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 $\nearrow$ Box 9 $\rightarrow$ Box 15 $\searrow$ Box 19
10	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 13 $\nearrow$ Box 10 $\rightarrow$ Box 16 $\searrow$ Box 18
11	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 7 $\nearrow$ Box 11 $\rightarrow$ Box 21 $\searrow$ Box 22
12	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 12 $\rightarrow$ Box 24
13	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 13 $\rightarrow$ Box 23
14	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 5 $\nearrow$ Box 14 $\rightarrow$ Box 24 $\searrow$ Box 27
Continued on the next page.		

Continuation of Table V ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
15	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 8 ↗ Box 15 → Box 20 ↘ Box 26
16	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 16 → Box 21
17	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 17 → Box 28
18	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 9 ↗ Box 15 → Box 23 ↘ Box 25
19	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 19 → Box 26
20	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 11 ↗ Box 20 → Box 12 ↘ Box 30
21	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 21 → Box 13
Continued on the next page.		

Continuation of Table V ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
22	$0 \leq x_1 \leq x_0$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 14 $\nearrow$ Box 22 $\rightarrow$ Box 17 $\searrow$ Box 31
23	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 23 $\rightarrow$ Box 12
24	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 24 $\rightarrow$ Box 19
25	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $0 \leq x_4 \leq x_0$ $x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 15 $\nearrow$ Box 25 $\rightarrow$ Box 16 $\searrow$ Box 29
26	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $0 \leq x_3 \leq x_0$ $x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$ $0 \leq x_5 \leq x_0$	Box 26 $\rightarrow$ Box 17
27	$0 \leq x_1 \leq x_0$ $x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 4 $\nearrow$ Box 27 $\rightarrow$ Box 13 $\searrow$ Box 32
28	$x_0 \leq x_1 \leq \frac{\alpha}{\gamma}$ $0 \leq x_2 \leq x_0$ $x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$ $0 \leq x_4 \leq x_0$ $0 \leq x_5 \leq x_0$	Box 28 $\rightarrow$ Box 16
Continued on the next page.		

Continuation of Table V ...		
Box	Coordinates of the hyperrectangle	Direction of the flow of vector field
29	$0 \leq x_1 \leq x_0$	Box 20
	$0 \leq x_2 \leq x_0$	$\nearrow$
	$0 \leq x_3 \leq x_0$	Box 29 $\rightarrow$ Box 21
	$0 \leq x_4 \leq x_0$	$\searrow$
	$x_0 \leq x_5 \leq \frac{\alpha}{\gamma}$	Box 23
30	$0 \leq x_1 \leq x_0$	Box 22
	$0 \leq x_2 \leq x_0$	$\nearrow$
	$0 \leq x_3 \leq x_0$	Box 30 $\rightarrow$ Box 24
	$x_0 \leq x_4 \leq \frac{\alpha}{\gamma}$	$\searrow$
	$0 \leq x_5 \leq x_0$	Box 26
31	$0 \leq x_1 \leq x_0$	Box 21
	$0 \leq x_2 \leq x_0$	$\nearrow$
	$x_0 \leq x_3 \leq \frac{\alpha}{\gamma}$	Box 31 $\rightarrow$ Box 27
	$0 \leq x_4 \leq x_0$	$\searrow$
	$0 \leq x_5 \leq x_0$	Box 28
32	$0 \leq x_1 \leq x_0$	Box 3
	$x_0 \leq x_2 \leq \frac{\alpha}{\gamma}$	$\nearrow$
	$0 \leq x_3 \leq x_0$	Box 32 $\rightarrow$ Box 23
	$0 \leq x_4 \leq x_0$	$\searrow$
	$0 \leq x_5 \leq x_0$	Box 24

Upon inspection of Table V, we find that the vector field formed a continuous map of a cross section onto itself. The continuous map is as follows:

$$\boxed{12 \rightarrow 24 \rightarrow 19 \rightarrow 26 \rightarrow 17 \rightarrow 28 \rightarrow 16 \rightarrow 21 \rightarrow 13 \rightarrow 23 \rightarrow 12}.$$

We obtained a toroidal-like manifold (which is a positively invariant set) and a cross section of this structure formed a continuous map onto itself. So, by Brouwer's Fixed Point theorem, there exists a limit cycle.

VI. LOCALIZATION OF LIMIT CYCLE

We extended the core idea of the Brouwer’s Fixed Point theorem to an interval-based Reachability Analysis [23] to localize the limit cycle trajectory. This localization is the natural next step in rigorously refining the region of existence of the limit cycle. This method is based on rigorously solving the network dynamics using Interval Analysis. Starting from a set of initial conditions, a hypercube is obtained that rigorously contains all trajectories. The sets that mapped onto themselves under the vector field of the system dynamics, rigorously computed using Reachability Analysis, contain a limit cycle. This is a direct application of the Brouwer’s Fixed Point theorem on refinement of the existence region.

This interval-based Reachability Analysis approach thus serves as an implementation of Brouwer’s Fixed Point theorem’s topological arguments using numerical techniques. While the theorem guarantees the presence of a fixed point and thus a periodic orbit, Reachability Analysis refines this conclusion by localizing the region where the limit cycle trajectory lies. In this approach, the geometric confinement test extends the theoretical proof by connecting the existence result to a rigorous characterization of the system’s oscillatory dynamics.

A. Methodology

The methodology involves dividing a search space into smaller hyperrectangles, evolving the system dynamics from each via interval-based Reachability Analysis (flowpipes), and examining the return map to classify each region as: (i) Completely contains the limit cycle, (ii) Partially contains the limit cycle, or (iii) Does not contain the limit cycle. This classification fundamentally depends on a geometric subset test to check if the computed return set, X_t , is entirely encompassed within the original set, X_0 .

To reduce the computational complexity of performing this subset test on 3D zonotopic sets, our approach projects the sets onto the three principal 2D planes (xy , yz , and xz). For each projection, a thorough `multi_line_check` is performed, which evaluates the 2D subset property by traversing directional 1D “slices” across the sets. The classification is thereafter done based on the outcomes of these evaluations:

- **Contains limit cycle completely** is assigned solely when an intersection is observed and all three 2D projection evaluations unanimously indicate “complete”.
- **Does not contain limit cycle** is assigned when a failure is observed. This arises if (i) at least one 2D projection verification exceeds the subset condition, or (ii) no observable intersection is identified and the bounding box of the return set does not intersect with that of the original set.

- **Partially contains limit cycle** is utilized in uncertain scenarios. This occurs if (i) an observable intersection is observed but the three-plane evaluation produces mixed outcomes (a combination of “complete” and “uncertain”), or (ii) no apparent intersection is recognized yet the resultant set represents a “near miss,” characterized by an overlap of its bounding box with that of the original set.
- **Needs further check** is assigned for purely computational failures, such as when the `nearest` function cannot identify a valid return candidate.

B. Algorithms

An external loop initially calls the function `divide`, to split the entire search space into smaller subsets. The `classify` function is thereafter applied iteratively to each subset, facilitating systematic identification and categorization and thereby localization of regions in the phase space that contain the limit cycle of the repressilator model.

The main function, `classify`, executes the core analysis for each initial set. The operation depends on several subroutines: (i) `solve_ivp`, which simulates the system’s dynamics, where the initial conditions are defined as interval sets generated by partitioning the phase plane into a grid via the `divide`; (ii) `nearest`, which determines the trajectory segment closest to the initial set; (iii) `classify_three_planes`, which enhances classification via geometric containment checks across various 2D projections; and (iv) `multi_line_check` and `hull_line`, which ensure accurate verification of containment boundaries.

The algorithms outlined here define the process for identifying and categorizing and thereby localizing limit cycle trajectory within the phase space of the repressilator.

1) Algorithm 1 - Overall Classification Procedure (Main Loop): This algorithm begins by dividing the entire search space into a grid of smaller, distinct boxes. The algorithm subsequently checks each of these boxes, executing the primary classification process to determine its correlation with the system’s limit cycle. According to the returning set, each box is classified into one of the three categories: total containment, partial containment, or requiring further investigation due to computational uncertainty. Boxes found not to contain the cycle are eliminated, so significantly reducing the search space.

Algorithm 1: Overall Classification Procedure (Main Loop)

Input: Search space X_{search} , number of divisions n_{div}

Output: Lists: $Limit_Cycle$, idx_limit , $Limit_Cycle_partial$, $idx_partial$, $Further_check$, $idx_further_check$, $Failed_Boxes$

▷ Initialize output lists.

```
1  $Limit\_Cycle \leftarrow []$ ,  $idx\_limit \leftarrow []$ ,  $Limit\_Cycle\_partial \leftarrow []$ ,  $idx\_partial \leftarrow []$ ,  
    $Further\_check \leftarrow []$ ,  $idx\_further\_check \leftarrow []$ ,  $Failed\_Boxes \leftarrow []$ 
```

```
2  $S \leftarrow \text{divide}(X_{search}, n_{div})$                                 ▷ Split the initial search space.
```

```
3 foreach  $(X_0, idx) \in \text{enumerate}(S)$  do
```

```
4    $status \leftarrow \text{classify}(X_0, idx)$ 
```

```
5   if  $status = \text{"Contains limit cycle completely"}$  then
```

```
6     append  $X_0$  to  $Limit\_Cycle$ 
```

```
7     append  $idx$  to  $idx\_limit$ 
```

```
8   else if  $status = \text{"Contains limit cycle partially"}$  then
```

```
9     append  $X_0$  to  $Limit\_Cycle\_partial$ 
```

```
10    append  $idx$  to  $idx\_partial$ 
```

```
11  else if  $status = \text{"Needs further check with stricter settings"}$  then
```

```
12    append  $X_0$  to  $Further\_check$ 
```

```
13    append  $idx$  to  $idx\_further\_check$ 
```

▷ Boxes that do not contain the cycle are discarded.

▷ Failed boxes are added to $Failed_Boxes$ inside solve_ivp .

2) *Algorithm 2 - classify*: This algorithm functions as the primary classification function for a singular initial set (X_0). Initially, it models the system's trajectory via the solve_ivp function. It then determines the point of closest return to X_0 utilizing the nearest function. The categorization proceeds according to the characteristics of the return set. If a valid geometric intersection is found, a rigorous multi-plane geometric verification is carried out. If just a "closest" return point is identified without an intersection, the algorithm employs a more cautious check. This check is done by evaluating the overlap of the bounding boxes of the initial and final sets. This approach involves error handling and guarantees reliable classification.

Algorithm 2: Main routine: `classify`

Input: Initial set X_0 (Hyperrectangle), box index idx , skip parameter s , directions N

Output: Classification string

```

1 try  $(\mathcal{S}, success) \leftarrow solve\_ivp(X_0)$ 
2 if not success then
3   return “solve_ivp failed”
4   append  $X_0$  to Failed_Boxes
5  $(k^*, status) \leftarrow nearest(\mathcal{S}, s)$ 
6 if  $status = :failed$  then
7   return “Needs further check with stricter settings”
8  $X_t \leftarrow set(\mathcal{S}[k^*])$ 
9  $X_t^{HR} \leftarrow overapproximate(X_t, Hyperrectangle)$ 
10 if  $status = :intersected$  then
11   return classify_three_planes( $X_0, X_t^{HR}, N$ )
12 else if  $status = :closest\_fallback$  then
13   if  $intersection(X_0, X_t^{HR}) \neq \emptyset$  then
14     return “Contains limit cycle partially”
15   else
16     return “Does not contain limit cycle”
17 catch any error return “Needs further check with stricter settings”

```

3) *Algorithm 3 - solve_ivp*: This algorithm numerically solves the initial value problem (IVP) for a specified initial set, X_0 . It employs the Taylor-model-based solver, TMJets21a, to calculate the evolution of the system’s dynamics over a predetermined time horizon. The resultant solution is contained within a series of zonotopes, creating a “flowpipe” that ensures the true trajectory remains contained. In case the solver encounters trouble with a specific initial set, the function records the set for subsequent examination and issues a failure flag, so allowing the entire analysis to continue uninterrupted.

Algorithm 3: `solve_ivp`

Input: Initial set X_0

Output: Solution flowpipe $\mathcal{S}$, success flag

```

1 try Define IVP with repressilator dynamics from  $X_0$ 
2 Solve using TMJets21a algorithm
3  $\mathcal{S} \leftarrow \text{overapproximate}(\text{solution}, \text{Zonotope})$ 
4 return ( $\mathcal{S}, \text{true}$ )
5 catch failure append  $X_0$  to Failed_Boxes
6 return (nothing, false)

```

4) *Algorithm 4 - nearest:* This algorithm identifies the system’s trajectory “return time” by examining the solution flowpipe. It utilizes a two-step approach to guarantee both precision and robustness. In the initial phase, it examines the time steps of the solution, finding the earliest geometric intersection with the beginning state. If an intersection is found, it provides a successful result. Simultaneously, it monitors the state that is geometrically nearest to the starting state, as determined by the distance between their centers. A skip parameter s is used to avoid the categorization of the initial time steps in the solution flowpipe as the nearest return set. If the initial step fails to identify a direct intersection, a more focused search is used within a limited region surrounding the nearest option. Should this also fail, the method defaults to returning the index of the “closest” state, so offering a heuristic-based outcome in the absence of a confirmed one. This dual methodology guarantees the identification of a significant return point, even in geometrically complex or numerically complex situations.

Algorithm 4: nearest

Input: Solution flowpipe $\mathcal{S}$, skip parameter s **Output:** Tuple (k^*, status)

```

1  $Z_0 \leftarrow \text{set}(\mathcal{S}[1]), c_0 \leftarrow \text{center}(Z_0)$ 
2  $\text{best\_dist} \leftarrow \infty, \text{best\_k} \leftarrow \text{nothing}$ 

  ▷ Step 1: Search for an intersection.
3 for  $k = s$  to  $\text{enumerate}(\mathcal{S})$  do
4    $Z_k \leftarrow \text{set}(\mathcal{S}[k])$ 
5   try if  $\text{intersection}(Z_k, Z_0) \neq \emptyset$  then
6     return  $(k, : \text{intersected})$ 
7   catch error                                     ▷ Ignore intersection check errors.
8    $d \leftarrow \|\text{center}(Z_k) - c_0\|$ 
9   if  $d < \text{best\_dist}$  then
10     $\text{best\_dist} \leftarrow d, \text{best\_k} \leftarrow k$ 

  ▷ Step 2: Fallback if no intersection was found.
11 if  $\text{best\_k} \neq \text{nothing}$  then
12    $\text{window} \leftarrow \max(1, \text{best\_k} - 5) \dots \min(\text{enumerate}(\mathcal{S}), \text{best\_k} + 5)$ 
13   foreach  $\tilde{k} \in \text{window}$  do
14     if  $\text{intersection}(\text{set}(\mathcal{S}[\tilde{k}]), Z_0) \neq \emptyset$  then
15       return  $(\tilde{k}, : \text{intersected})$ 
16   return  $(\text{best\_k}, : \text{closest\_fallback})$ 
17 return  $(1, : \text{failed})$ 

```

5) *Algorithm 5 - classify_three_planes*: This algorithm confirms whether a 3D return set (X_t) is contained within its matching 3D initial set (X_0). The algorithm simplifies the problem by projecting both hyperrectangles onto the three primary 2D coordinate planes: xy , yz , and xz , rather than executing a complex 3D comparison. For each of these two-dimensional projections, a containment check is conducted with the `multi_line_check` function. The final classification of “completely contained”, “partially contained”, or “not contained” is made based on the cumulative outcomes from the three planes, guaranteeing a dependable conclusion even when one of the 2D assessments may be uncertain.

Algorithm 5: `classify_three_planes`

Input: Sets X_0, X_t (Hyperrectangles), directions N
Output: Classification

```

1 complete  $\leftarrow 0$ , not_complete  $\leftarrow 0$ , uncertain  $\leftarrow 0$ 
2 foreach plane  $(i, j) \in \{(1, 2), (2, 3), (1, 3)\}$  do
3    $X_0^p \leftarrow \text{project}(X_0, [i, j])$ ,  $X_t^p \leftarrow \text{project}(X_t, [i, j])$ 
4   if  $\| \text{center}(X_0^p) - \text{center}(X_t^p) \| < \text{tolerance}$  then
5     complete  $\leftarrow \text{complete} + 1$ 
6     continue
7   result  $\leftarrow \text{multi\_line\_check}(X_0^p, X_t^p, N)$ 
8   if result = :complete then
9     complete  $\leftarrow \text{complete} + 1$ 
10  else if result = :not_complete then
11    not_complete  $\leftarrow \text{not\_complete} + 1$ 
12  else
13    uncertain  $\leftarrow \text{uncertain} + 1$ 

  ▷ Final decision based on results from the three planes.
14 if not_complete  $> 0$  then
15   return “Does not contain limit cycle”
16 else if complete == 3 then
17   return “Contains limit cycle completely”
18 else
19   return “Contains limit cycle partially”

```

6) *Algorithm 6 - multi_line_check*: This algorithm conducts a containment check between two 2D sets by probing them from multiple directions. For each of the N chosen directions, it first performs an exact geometric check by comparing the line segments corresponding to the intersections of the two sets along that direction. If the primary geometric test fails because of numerical instability or nonconvex boundary behavior, the algorithm switches to a more robust backup method that uses support functions to estimate the maximum extent of each set along the same direction.

The support function, denoted by $\rho(u, X)$, gives the farthest distance of set X in the direction of the unit vector u . By comparing $\rho(u, X_t^P)$ and $\rho(u, X_0^P)$ and their opposites, $\rho(-u, X_t^P)$ and $\rho(-u, X_0^P)$, the algorithm can determine whether the evolved set X_t^P extends beyond the initial set X_0^P along any direction u . If this occurs, geometric containment is refuted. This methodology guarantees reliability: if any direction conclusively shows that X_t^P exceeds X_0^P , containment is immediately rejected. If all tests pass, containment is considered complete. In case any numerical errors occur, the result is conservatively marked as uncertain, avoiding false positives.

Algorithm 6: multi_line_check

Input: 2D projections X_0^P, X_t^P , number of directions N

Output: Containment status symbol

```

1  Generate  $N$  unit directions  $\{u_k\}$ 
2   $fail\_count \leftarrow 0$ ,  $uncertain \leftarrow \text{false}$ 
3  foreach  $u_k$  in directions do
4       $success \leftarrow \text{false}$ 
        ▷ Primary Method: Geometric check.
5      try  $l \leftarrow \text{hull\_line}(X_0^P, u_k)$ 
6       $inter_0 \leftarrow l \cap X_0^P$ ,  $inter_t \leftarrow l \cap X_t^P$ 
7      if  $inter_t \not\subseteq inter_0$  then
8          return :not_complete
9       $success \leftarrow \text{true}$ 
10     catch error  $fail\_count \leftarrow fail\_count + 1$ 
        ▷ Fallback Method: Support function check.
11     if not success then
12         try if  $\rho(u_k, X_t^P) > \rho(u_k, X_0^P)$  or  $\rho(-u_k, X_t^P) > \rho(-u_k, X_0^P)$  then
13             return :not_complete
14         catch error  $uncertain \leftarrow \text{true}$ 
15 if  $fail\_count/N > \text{max\_fail\_fraction}$  or  $uncertain$  then
16     return :uncertain
17 else
18     return :complete

```

7) *Algorithm 7 - hull_line*: This algorithm calculates a line segment that indicates the overall extent, or “width” of a 2D geometric set in a designated direction. It achieves this by identifying the two most extreme points on the boundary of the set utilizing the support function σ , which determines the points that are farthest in the specified direction and its opposite. The function includes a fallback mechanism to guarantee robustness against numerical complexities or complex geometries. Should the support function fail on the original set, it then calculates a more straightforward hyperrectangle enclosure (a bounding box) and subsequently identifies the extreme points on that simplified shape. This ensures that a valid line segment is consistently produced.

Algorithm 7: hull_line

Input: 2D set Z , direction u

Output: Line segment connecting extreme points

```

1  $d \leftarrow u / \|u\|$ 
2 try  $p \leftarrow \sigma(d, Z)$                                  $\triangleright$  Support point in direction  $d$ .
3  $q \leftarrow \sigma(-d, Z)$                                  $\triangleright$  Support point in direction  $-d$ .
4 return LineSegment( $p, q$ )
5 catch error       $\triangleright$  Fallback to a bounding box if support vector fails.
6  $HR \leftarrow \text{overapproximate}(Z, \text{Hyperrectangle})$ 
7  $p \leftarrow \sigma(d, HR)$ 
8  $q \leftarrow \sigma(-d, HR)$ 
9 return LineSegment( $p, q$ )
```

C. Results and Discussions

The existence proof for the three-gene cyclic regulatory network within an invariant cube $0 \leq x_1, x_2, x_3 \leq \frac{\alpha}{\gamma}$, when divided into 2^3 smaller boxes by the surfaces $x_i = x_0$ (for $i = 1, 2, 3$), gives us six cross-sections of the smaller boxes that map onto themselves. We then illustrated the interval-based Reachability Analysis to localize periodic orbit through one of these cross-sections that map onto themselves in the three-gene cyclic regulatory network. The framework successfully divided the cross-section according to the underlying dynamical behaviour of the three-gene cyclic regulatory network. The methodology, based on partitioning the cross-section and analyzing the flowpipe of each partition, showed several findings, which are visually summarized in Fig. 1

The analysis classified the vast majority of the partitioned region of this cross-section as not containing a periodic orbit. Region a , small_X0[1191] in Figure 1, contains the unstable steady state

(x_0, x_0, x_0) (the value of x_0 for this simulation is 5.361) of the system and is excluded from the Reachability Analysis. Region b , `small_X0[1121, 1156]`, shows the classification of boxes near the excluded steady-state point as “Completely Contains Cycle”. This result is physically counterintuitive, as these trajectories lie deep within the steady-state basin of attraction and should converge to it. For trajectories initialized in the basin of the steady state, the reachable sets contract monotonically as they approach equilibrium. Consequently, the final contracted overapproximation can become geometrically contained within the initial, larger hyperrectangle, satisfying the inclusion criterion despite the absence of periodic behaviour. A distinct cluster of initial sets Region c , `small_X0[793:795, 828:830, 865]`, in Figure 1, was classified as “Completely Contains Cycle” or “Partially Contains Cycle”. This region is located far from the system’s unstable steady state which is contained in the Region a . Further in Figure 2 the numerical solution of the repressilator model is plotted with one of the cross-section that maps onto itself. Here the trajectory passes through the Region c in Figure 1, which was classified by the algorithm as containing the limit cycle completely or partially. Fig. 3 provides a magnified view of the trajectory passing through the region classified by the algorithm as containing the limit cycle completely or partially. This finding validates the method’s efficacy in localizing the region in which the limit cycle exists.

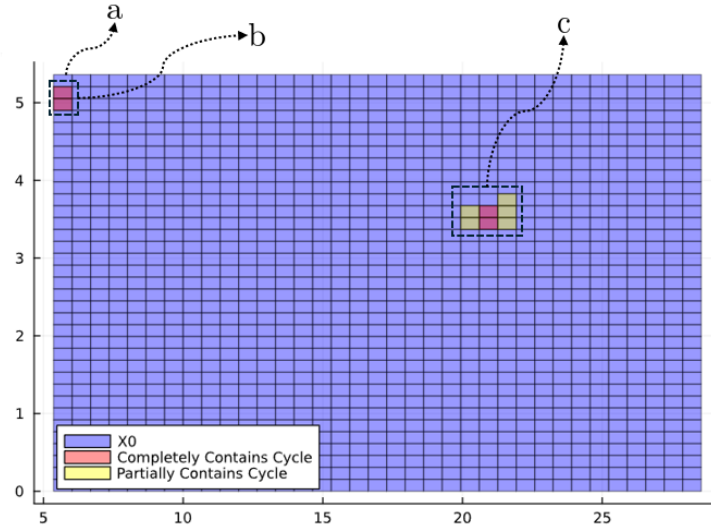


Figure 1: xy -projection of the cross-section, that maps onto itself, showing the geometric classification results. Region a (`small_X0[1191]`) contains the steady-state point and was excluded from the reachability analysis. Region b (`small_X0[1121, 1156]`) shows a false “Completely Contains Cycle” classification arising from convergence near the steady state. Region c (`small_X0[793:795, 828:830, 865]`) corresponds to the cluster of boxes through which the limit cycle passes completely or partially.

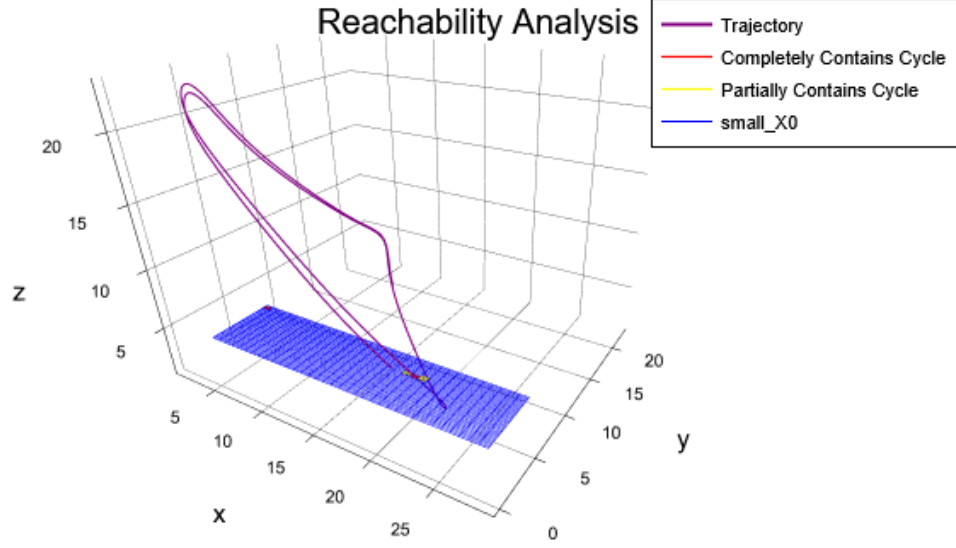


Figure 2: 3D plot of the numerical solution of the repressilator model with initial condition $(19, 3.5, 5.361)$. The resulting trajectory passes through the region c of Figure 1. The system parameters used in the simulation are: $\alpha = 26.26$, $\gamma = 0.92$, $k = 4$, $n = 5$. x, y, z axes represent the concentration of i -th (for $i = 1, 2, 3$) protein respectively.

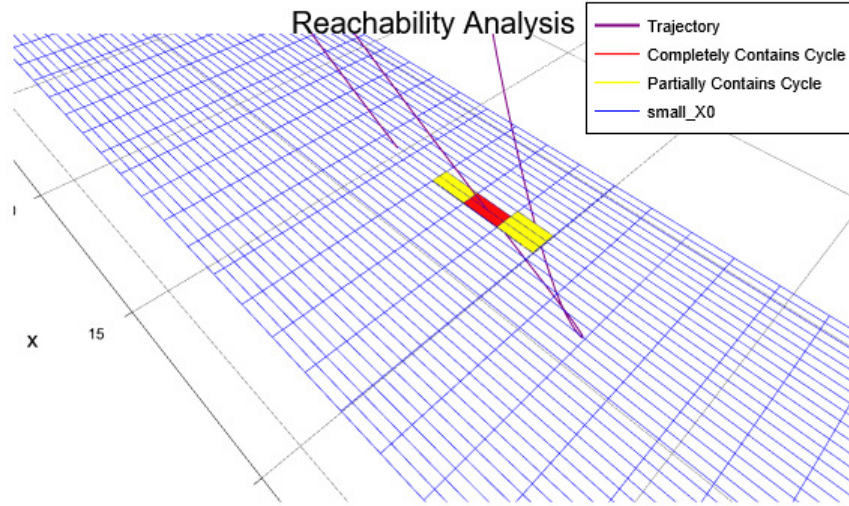


Figure 3: A magnified view of Figure 2 showing the intersection of the numerical solution with the cross-section.

VII. CONCLUSION

We proved the existence of a limit cycle in a class of cyclic gene regulatory networks and provided a rigorous localization of this region of existence. The existence proof, based on Brouwer's Fixed Point theorem, is an alternative and elementary proof for this class of systems, with the benefit of an enhanced geometric visualisation. This rigorous localization relies on an interval-based Reachability Analysis, which, in conjunction with Brouwer's Fixed Point theorem, can provide a significantly tighter enclosure of the limit cycle trajectory.

For the existence proof, we constructed a positively invariant set for the network dynamics by exploiting the inherent boundedness and by creating a toroidal-like manifold region that eliminates hypervolume consisting of the unstable steady state and its associated stable manifolds. We found that a cross section of this manifold continuously maps onto itself under the system dynamics, showing via Brouwer's Fixed Point theorem, the existence of a limit cycle. These results are illustrated using a 5-dimensional example. After establishing the existence of a limit cycle in cyclic gene regulatory networks, we employed interval-based Reachability Analysis to localize the region of its existence across the cross-sections that map onto themselves. The Reachability Analysis was demonstrated using a 3-dimensional example, offering a visual representation of the categorized regions.

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